

ResearchSwarm

Multi-Agent Deep Research Engine

"Deeper research. Faster insights. Powered by a swarm."

Microsoft Build AI Hackathon — Agent Swarms Track

The Problem

- Researching complex topics takes hours of manual work
- Single LLM calls produce shallow, unverified answers
- No visibility into sources or reasoning
- Critical gaps and hallucinations go undetected
- Knowledge workers spend 30% of their time just gathering information

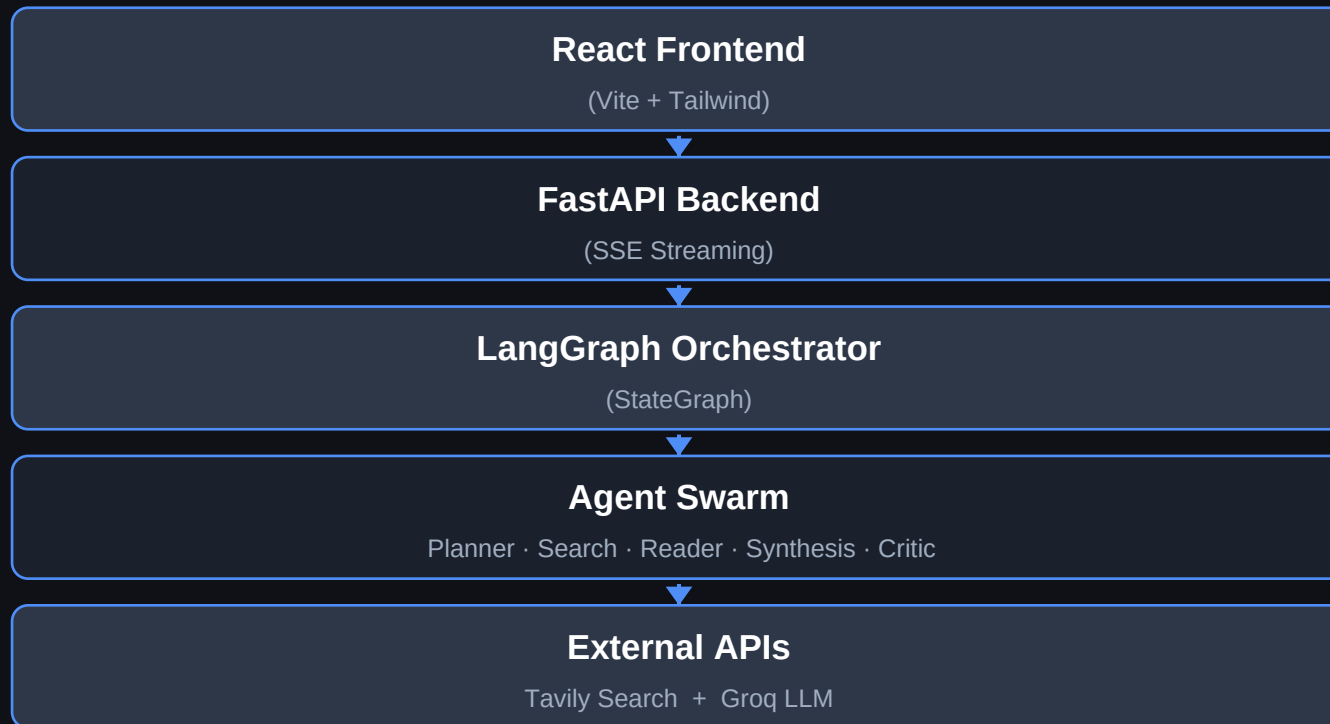
The Solution

- ResearchSwarm deploys a coordinated swarm of specialized AI agents
- Each agent has one job and does it in parallel
- Result: comprehensive, cited, verified research reports in minutes
- Full source transparency and real-time agent visibility
- A Critic Agent scores confidence and flags unsupported claims

How It Works



Architecture



AI Integration

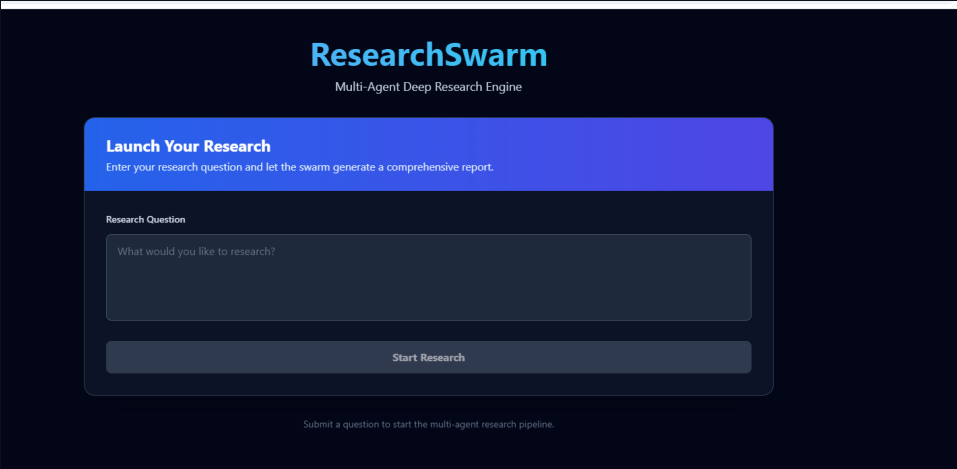
Models & APIs

- **Planner Agent:** Groq Llama 3.1 8b (fast)
- **Search Agent:** Tavily Search API
- **Reader Agent:** Groq Llama 3.1 8b (fast)
- **Synthesis Agent:** Groq Llama 3.3 70b (powerful)
- **Critic Agent:** Groq Llama 3.3 70b (powerful)

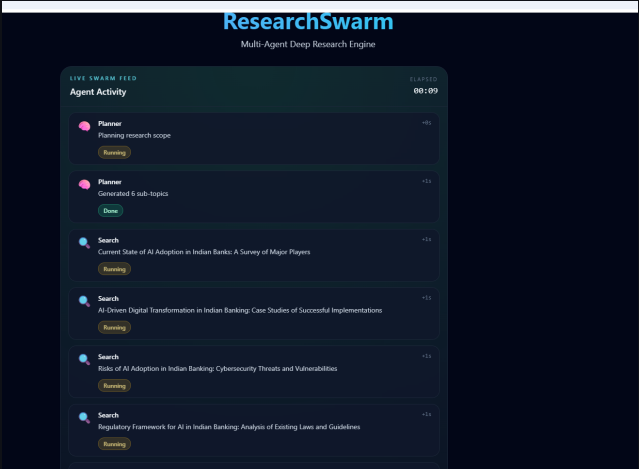
Framework & Stack

- **Orchestration:** LangGraph
- **Parallelism:** asyncio
- **Streaming:** FastAPI SSE
- **Frontend:** React + Vite
- **Deployment:** Render + Vercel

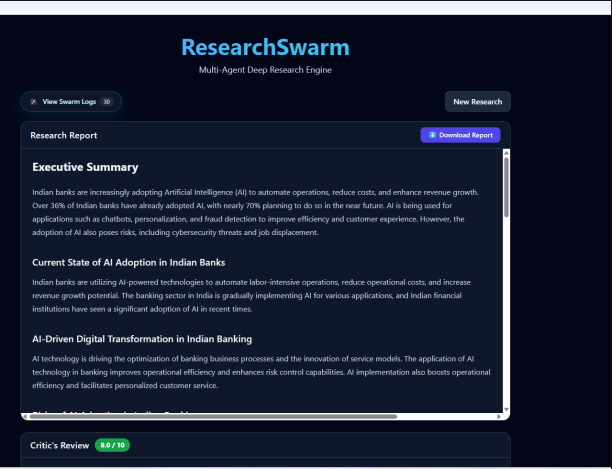
Live Demo



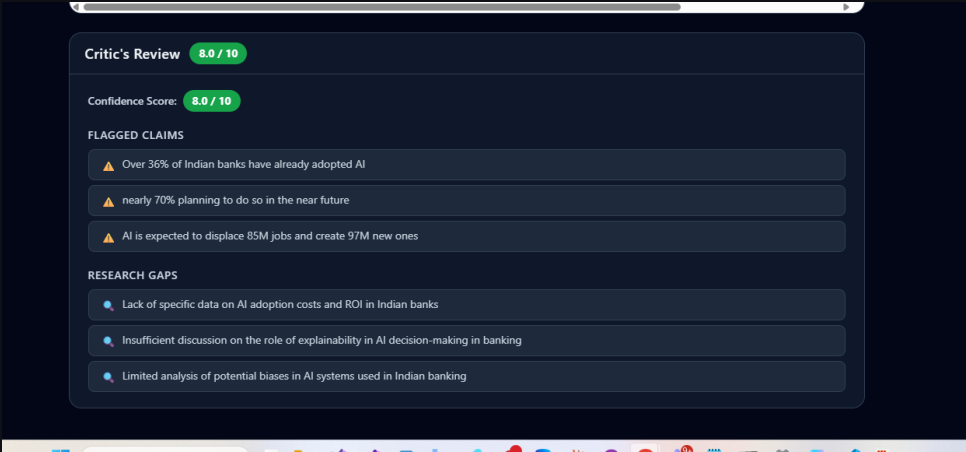
Launch Interface



Live Agent Feed



Research Report



Report Continued

Key Features

- Real-time agent progress feed
- Parallel multi-agent execution
- Streaming report — no waiting for full pipeline
- Critic reviews with confidence scoring
- Full source citations
- Deployed and live — not just a prototype

Impact & Use Cases

Who benefits

- Researchers and analysts
- Journalists and writers
- Students and academics
- Business strategy teams

Real-world applications

- Competitive intelligence
- Market research
- Due diligence
- Policy research

The Team

Participant:	Rupam Pal
Background:	Full Stack Developer
Hackathon:	Microsoft Build AI Hackathon
Track:	Agent Swarms (Theme 05)
GitHub:	https://github.com/Rupam0710/-ResearchSwarm-Multi-Agent-Deep-Research-Engine
Live Demo:	https://research-swarm-multi-agent-deep-res.vercel.app/